Replay-Aware TSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: tsplib_no_replay.
- Best TSPLIB holdout gap: tsplib_no_replay.
- Best synthetic holdout gap: tsplib_no_replay.

Run Metadata

- run_name: run_20260513_152340_q
- started_at_local: 2026-05-13 15:23:40
- finished_at_local: 2026-05-13 15:33:45
- duration_hhmm: 00:10
- duration_seconds: 604.771
- seed_offset: 16000
- replicate_label: q
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
tsplib_no_replay	none	score_only	False	0.078414	0.0	0.0499	0.676264	0.76	0.08058
tsplib_random_replay	random	score_only	False	0.097539	0.0	0.06207	0.623686	0.76	0.03663
tsplib_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.913641	0.56	-0.027259
tsplib_failure_replay_compression	failure	novelty_gate	True	0.082055	0.010343	0.055978	0.597264	0.76	0.046972

Condition Notes

tsplib_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.078414, synthetic 0.0, combined 0.0499.

- Accepted-epoch count 3, mean accepted code novelty 0.676264, and final complexity 0.76.
- Adaptation efficiency 0.08058 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.078414 across 7 instances; family means: ch=0.077853, kroD=0.096271, pcb=0.182953, pr=0.056565, rd=0.036662, st=0.020741.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 9644, "family": "berlin", "name": "berlin52", "optimality_gap": 0.278706}, {"best_known_cost": 2579, "cost": 3294, "family": "a", "name": "a280", "optimality_gap": 0.277239}, {"best_known_cost": 14379, "cost": 17216, "family": "lin", "name": "lin105", "optimality_gap": 0.197302}]

tsplib_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.097539, synthetic 0.0, combined 0.06207.
- Accepted-epoch count 3, mean accepted code novelty 0.623686, and final complexity 0.76.
- Adaptation efficiency 0.03663 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.097539 across 7 instances; family means: ch=0.050092, kroD=0.074058, pcb=0.189255, pr=0.112871, rd=0.130847, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3174, "family": "a", "name": "a280", "optimality_gap": 0.23071}, {"best_known_cost": 629, "cost": 684, "family": "eil", "name": "eil101", "optimality_gap": 0.08744}, {"best_known_cost": 14379, "cost": 15601, "family": "lin", "name": "lin105", "optimality_gap": 0.084985}]

tsplib_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.913641, and final complexity 0.56.
- Adaptation efficiency -0.027259 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "family": "a", "name": "a280", "optimality_gap": 0.401318}, {"best_known_cost": 629, "cost": 875, "family": "eil", "name": "eil101", "optimality_gap": 0.391097}, {"best_known_cost": 14379, "cost": 19107, "family": "lin", "name": "lin105", "optimality_gap": 0.328813}]

tsplib_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.082055, synthetic 0.010343, combined 0.055978.
- Accepted-epoch count 3, mean accepted code novelty 0.597264, and final complexity 0.76.
- Adaptation efficiency 0.046972 and archive sizes {'worst_cases': 6, 'failure_cases': 6, 'adversarial_layouts': 4} / elite 3.

- Panel heldout_tsplib mean gap 0.082055 across 7 instances; family means: ch=0.061264, kroD=0.060346, pcb=0.206881, pr=0.037704, rd=0.084703, st=0.062222.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3224, "family": "a", "name": "a280", "optimality_gap": 0.250097}, {"best_known_cost": 14379, "cost": 15729, "family": "lin", "name": "lin105", "optimality_gap": 0.093887}, {"best_known_cost": 7542, "cost": 7964, "family": "berlin", "name": "berlin52", "optimality_gap": 0.055953}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Suite Results

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (Mean)	Complexity	Notes
tsplib_no_replay	None	**0.078414**	**0.0**	**0.0499**	0.676	0.76	Best transfer and optimality gaps. No replay yields lowest gaps despite moderate code novelty.
tsplib_random_replay	Random	0.097539	0.0	0.06207	0.624	0.76	Slightly worse transfer and TSPLIB gaps with lower code novelty than no_replay.
tsplib_failure_replay	Failure	0.213387	0.064916	0.159398	**0.914**	0.56	Substantially worse transfer and TSPLIB gaps despite highest code novelty; no transfer benefit.
tsplib_failure_replay_compression	Failure + Compression	0.082055	0.010343	0.055978	0.597	0.76	Moderate gaps, better than failure_replay alone; code novelty falls while transfer improves.

Key Interpretations

- **Transfer and Optimality Gaps as Primary Indicators**

The **no replay** condition achieves the best held-out TSPLIB gap (0.0784) and synthetic gap (0.0), indicating superior transfer across problem families. The mean transfer gap is lowest (0.0499), confirming the best generalization. This condition should be considered the benchmark for transfer performance.

- **Effect of Replay on Transfer Performance**
- **Random replay** results in slightly degraded transfer and TSPLIB gaps compared to no replay; code novelty also decreases.
- **Failure replay without compression** yields the worst transfer and TSPLIB gaps despite showing the highest code novelty (0.91 mean). This suggests that increased code novelty does *not* translate to improved transfer or optimality, likely reflecting overfitting or ineffective replay dynamics.
- **Failure replay with compression** partly recovers transfer performance and optimality gaps (close to no replay) while code novelty drops significantly (to 0.60 mean), indicating compression modulates code novelty and supports better generalization.

- **Code Novelty vs. Transfer**

High code novelty under failure replay conditions does not correspond with better transfer or solution quality; in fact, the reverse is true. Conversely, the no replay condition maintains moderate novelty but achieves the best transfer metrics. This demonstrates that lexical novelty alone is an insufficient proxy for algorithmic invention without improvements in deterministic metrics.

- **Replay Modes Distinction**

The **no replay** mode outperforms **random replay** and **failure replay** modes clearly in transfer and held-out TSPLIB gaps. Failure replay results are particularly poor unless combined with compression-aware replay, which partly restores performance by reducing code novelty and maintaining complexity.

- **Complexity and Behavior**

Complexity is generally stable at 0.76 except for failure replay (0.56). Behavior profiles differ by replay mode but do not align simplistically with transfer performance—balanced profile in no replay condition has best results.

Conclusion

- The **no replay** condition demonstrates superior transfer to held-out TSPLIB and synthetic benchmarks, achieving the lowest optimality gaps and transfer gaps.
- Replay strategies (random, failure) degrade transfer and solution quality, despite failure replay increasing code novelty substantially.
- Compression-aware failure replay reduces code novelty and partially improves transfer metrics, though it does not surpass no replay.
- Lexical/code novelty elevation alone does not imply effective algorithmic invention without accompanying improvements in transfer or optimality gaps.
- Therefore, from a conservative, metric-prioritized perspective, **no replay is the most effective condition**, and replay methods require further refinement to improve transfer rather than merely increase novelty.